

7.5 inch
E-paper Display Series

GooDisplay®

GDEY075T7

Dalian Good Display Co., Ltd.

Product Specifications

Customer	**Standard**
Description	**7.5” E-PAPER DISPLAY**
Model Name	**GDEY075T7**
Date	**2021/03/10**
Revision	**1.0**

	Design Engineering		
	Approval	**Check**	**Design**
			

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1.Over View

GDEY075T7 is an Active Matrix Electrophoretic Display (AMEPD), with interface and a reference system design. The7.5” active area contains 800×480 pixels, and has 1-bit B/W full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC, SRAM, LUT, VCOM and border are supplied with each panel.

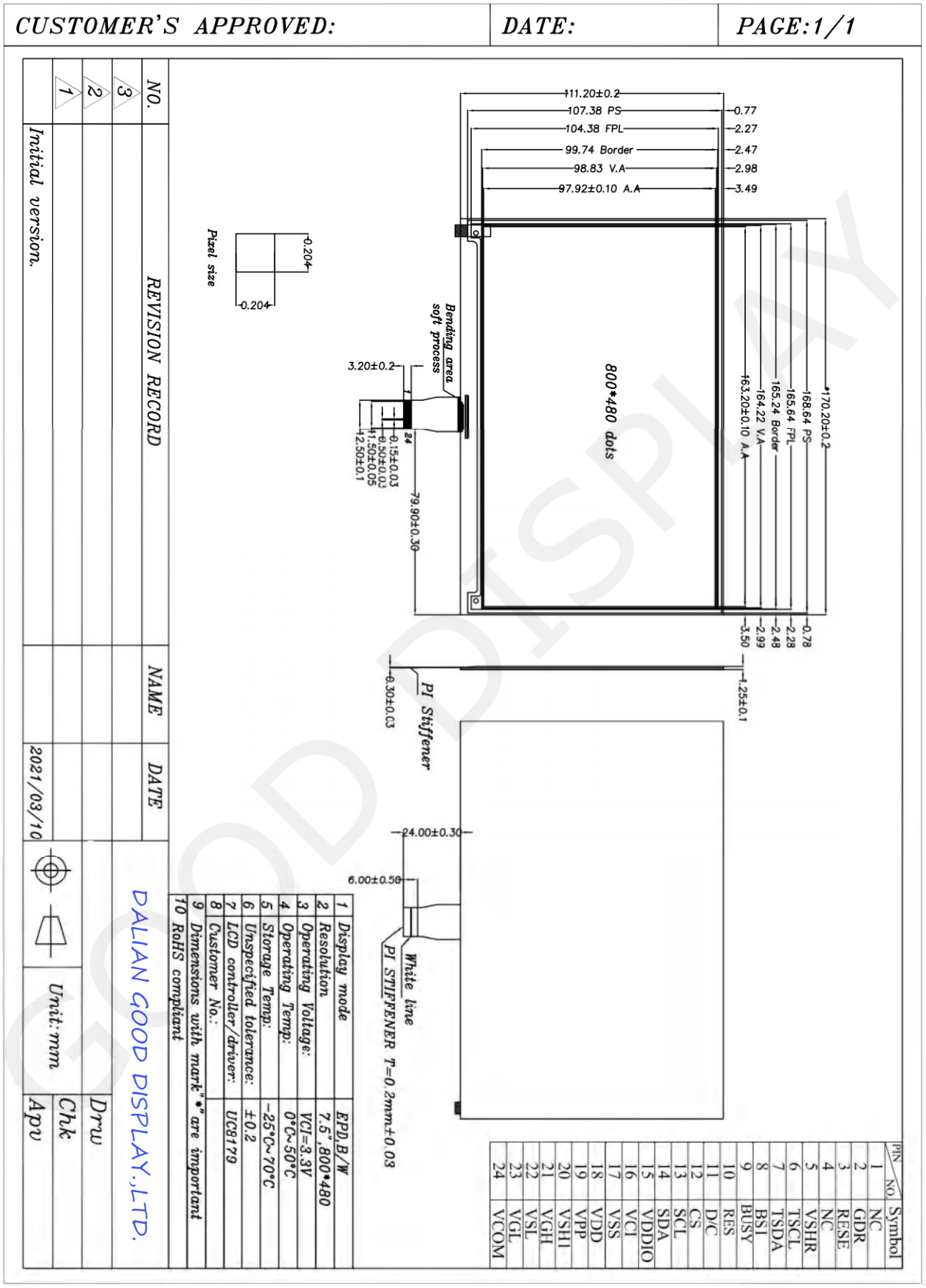
2.Features

- 800×480 pixels display
- High contrast
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Waveform stored in flash memory
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM,Gate and
- Source driving voltage
- I2C Signal Master Interface to read external temperature sensor/ built-in temperature sensor
- Available in COG package IC thickness 300um

3. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	7.5	Inch	
Display Resolution	800 (H)×480(V)	Pixel	Dpi:124
Active Area	163.20(H)×97.92(V)	mm	
Pixel Pitch	0.204×0.204	mm	
Pixel Configuration	Square		
Outline Dimension	170.20(H)×111.20 (V) ×1.25 (D)	mm	
Weight	44±0.5	g	

4. Mechanical Drawing of EPD module



5. Input /Output Pin Assignment

No.	Name	I/O	Description	Remark
1	NC		Do not connect with other NC pins	Keep Open
2	GDR	O	N-Channel MOSFET Gate Drive Control	
3	RESE	I	Current Sense Input for the Control Loop	
4	NC	NC	Do not connect with other NC pins	Keep Open
5	VSHR	C	Positive Source driving voltage(Red)	
6	TSCL	O	I^2C Interface to digital temperature sensor Clock pin	
7	TSDA	I/O	I^2C Interface to digital temperature sensor Data pin	
8	BS1	I	Bus Interface selection pin	Note 5-5
9	BUSY	O	Busy state output pin	Note 5-4
10	RES	I	Reset signal input. Active Low.	Note 5-3
11	D/C	I	Data /Command control pin	Note 5-2
12	CS	I	Chip select input pin	Note 5-1
13	SCL	I	Serial Clock pin (SPI)	
14	SDA	I/O	Serial Data pin (SPI)	
15	VDDIO	P	Power Supply for interface logic pins It should be connected with VCI	
16	VCI	P	Power Supply for the chip	
17	VSS	P	Ground	
18	VDD	C	Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS	
19	VPP	P	FOR TEST	
20	VSH1	C	Positive Source driving voltage	
21	VGH	C	Power Supply pin for Positive Gate driving voltage and VSH1	
22	VSL	C	Negative Source driving voltage	
23	VGL	C	Power Supply pin for Negative Gate driving voltage VCOM and VSL	
24	VCOM	C	VCOM driving voltage	

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I = Input Pin, O =Output Pin, I/O = Bi-directional Pin (Input/output), P = Power Pin, C =Capacitor Pin

Note 5-1: This pin (CSB) is the chip select input connecting to the MCU. The chip is enabled for MCU communication only when CSB is pulled Low.

Note 5-2: This pin (DC) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled Low, the data will be interpreted as command.

Note 5-3: This pin (RST) is reset signal input. The Reset is active Low
Note 5-4: This pin (BUSY) is BUSY state output pin. When BUSY is low, the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put BUSY pin low when the driver IC is working such as:

Outputting display waveform; or

Programming with OTP

Communicating with digital temperature sensor

BS1 State	MCU Interface
L	4-lines serial peripheral interface(SPI)
H	3- lines serial peripheral interface(SPI) - 9 bits SPI

Table: Bus interface selection

6.Electrical Characteristics

6.1 Absolute Maximum Rating

Parameter	Symbol	Rating	Unit
Logic supply voltage	VCI	-0.3 to +6.0	V
Logic Input voltage	VIN	-0.3 to VCI +0.3	V
Operating Temp range	TOPR	0 to +50	°C
Storage Temp range	TSTG	-25 to+70	°C
Optimal Storage Temp	TSTGo	23±2	°C
Optimal Storage Humidity	HSTGo	55±10	%RH

Note:

Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables

6.2 DC Characteristics

The following specifications apply for: VSS=0V, VCI=3.0V, TOPR =23ºC

Parameter	Symbol	Conditions	Applicable pin	Min.	Typ.	Max	Units
Single ground	V_{SS}	-		-	0	-	V
Logic supply voltage	V_{CI}	-	VCI	2.5	3.3	3.6	V
Core logic voltage	V_{DD}		VDD	2.5	3.3	3.6	V
High level input voltage	V_{IH}	-	-	0.7 V_{CI}	-	-	V
Low level input voltage	V_{IL}	-	-	-	-	0.3 V_{CI}	V
High level output voltage	V_{OH}	IOH = -400uA	-	VCI-0.4	-	-	V
Low level output voltage	V_{OL}	IOL = 400uA	-	0	-	0.4	V
Typical operating current	Iopr_V_{CI}	V_{CI} =3.3V	-	-	8	12	mA
Image update time	-	25 ºC	-	-	3	-	sec
Sleep mode current	Islp_V_{CI}	DC/DC off No clock No input load Ram data retain	-	-	25		uA
Deep sleep mode current	Idslp_V_{CI}	DC/DC off No clock No input load Ram data not retain	-	-	2	5	uA

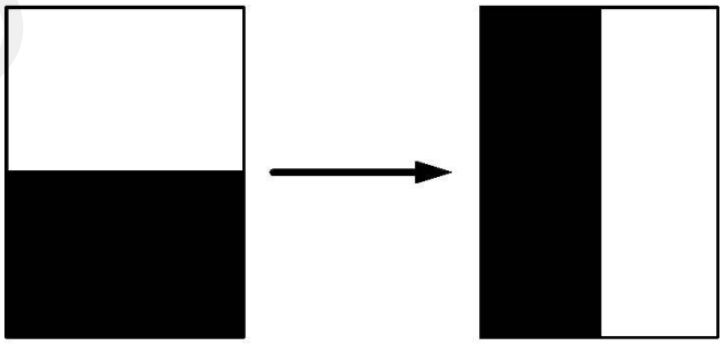
Notes:

1. The typical power is measured with following transition from horizontal 2 scale pattern to

vertical 2 scale pattern.

2.The deep sleep power is the consumed power when the panel controller is in deep sleep mode.

3.The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by GOOD DISPLAY



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6.3 AC Characteristics

6.3.1 MCU Interface Selection

Provides 3-wire/4-wire serial interface for command and display data transferred from the MCU. The serial interface supports 8-bit mode. Data can be input/output by clocks while the chip is active (CSB =LOW). While input, data are written in order from MSB at the clock rising edge. When too many parameters are input, the chip accepts only defined parameters, and ignores undefined ones.

BS	Interface	CSB	DC	SCL	SDA
High	3-wire SPI	Available	Fix to GND	Available	Available
Low	4-wire SPI	Available	Available	Available	Available

6.3.2 3 wire SPI format.

Data / Command is recognized with the first bit transferred. Data are transferred in the unit of 9 bits. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 9 bits. (The serial counter is reset at the rising edge of the CSB signal.)

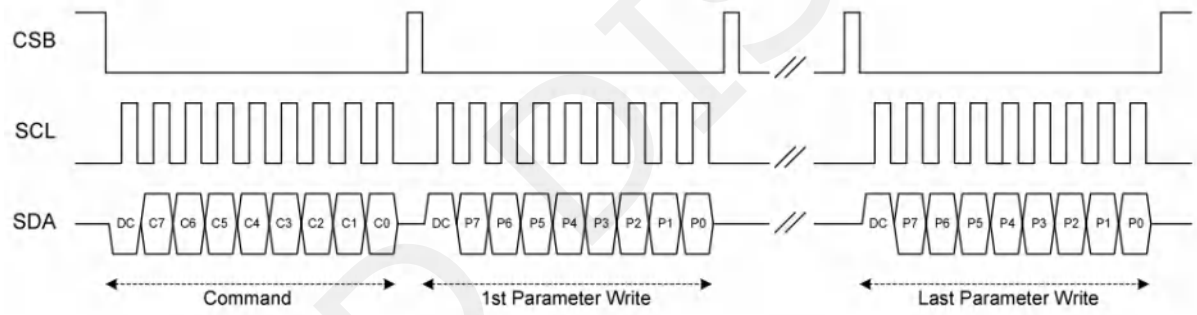


Figure: 3-wire SPI write operation

The MSB bit of data will be output at SDA pin after the 1st SCL falling edge, if the 1st input data at SDA is high.

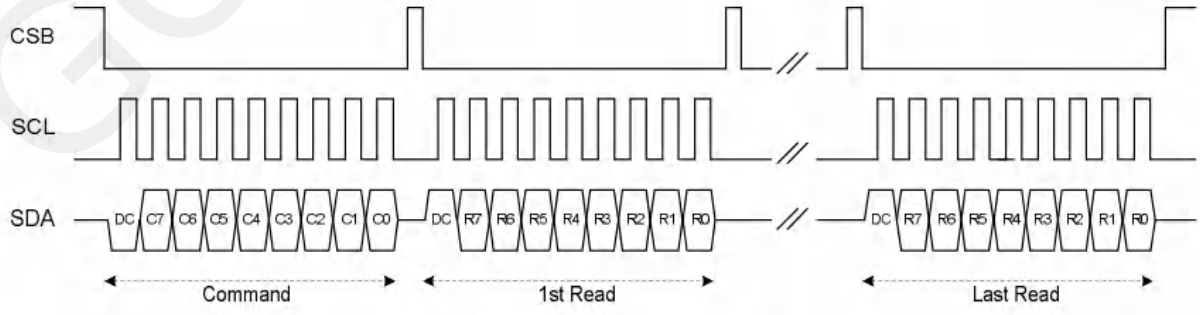


Figure: 3-wire SPI read operation

6.3.3 4 wire SPI format

Data / Command is recognized with DC pin. Data are transferred in the unit of 8 bits. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 8 bits. (The serial counter is reset at the rising edge of the CSB signal.)

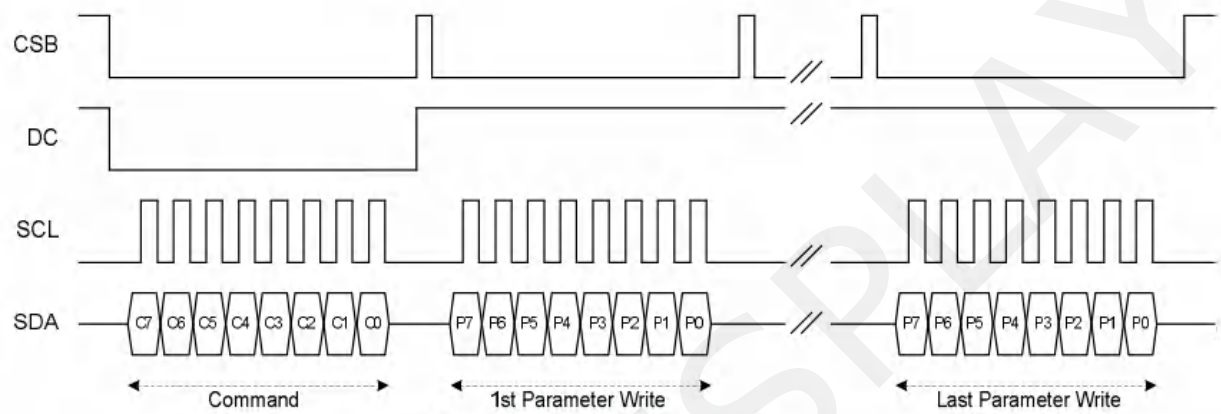


Figure: 4-wire SPI write operation

The MSB bit of data will be output at SDA pin after the CSB falling edge, if DC pin is High.

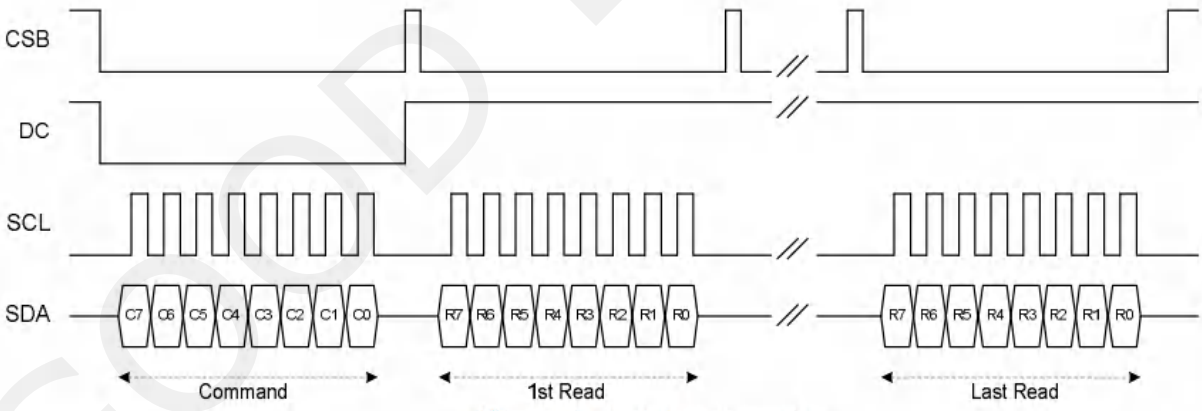


Figure: 4-wire SPI read operation

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6.3.4 3 wire dual SPI format.

Data / Command is recognized with the first bit transferred at SDA. Data are transferred in the unit of 5 SPI clocks. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 5 SPI clocks. (The serial counter is reset at the rising edge of the CSB signal.) In 3-wire dual SPI mode, SDA and SDA1 are only input mode for data write transmission.

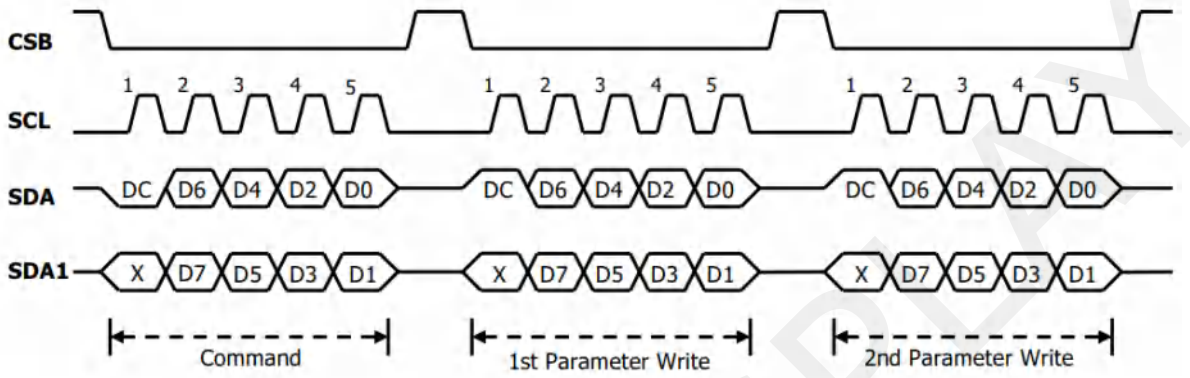


Figure: 3-wire dual SPI write operation

6.3.5 4 wire dual SPI format.

Data / Command is recognized with DC pin. Data are transferred in the unit of 4 SPI clocks. To prevent malfunction due to noise, it is recommended to set the CSB signal to HIGH every 4 SPI clocks. (The serial counter is reset at the rising edge of the CSB signal.) In 4-wire dual SPI mode, SDA and SDA1 are only input mode for data write transmission.

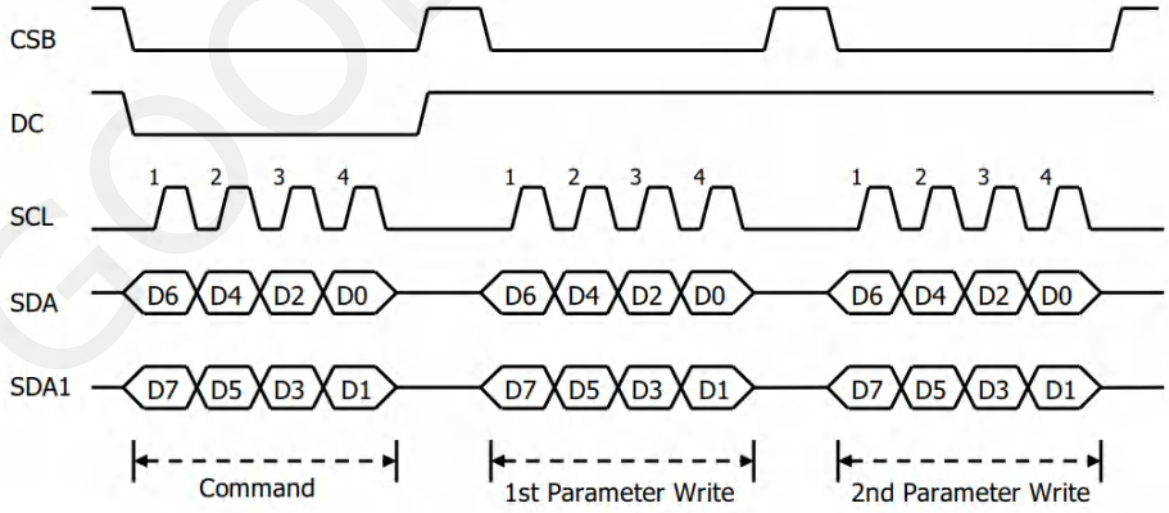
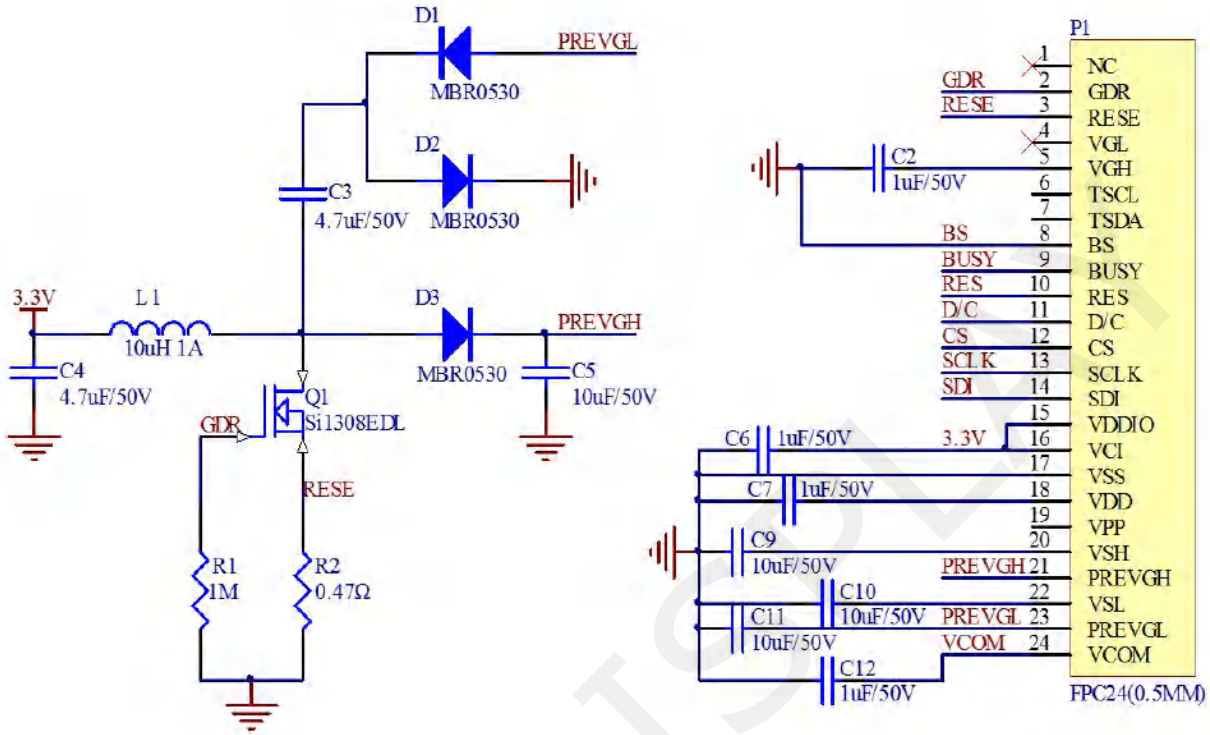


Figure: 4-wire dual SPI write operation

11. Reference Circuit



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